

MODULE 4 – SOLID AND HAZARDOUS WASTE MANAGEMENT

SOLID WASTE

Solid waste is the unwanted or useless solid materials generated from human activities in residential, industrial or commercial areas. Solid waste comprises of all the wastes arising from human and animal activities that are typically solid. It is all-inclusive of the heterogeneous mass from the urban community, as well as a more homogeneous accumulation of agriculture and industrial wastes.

Solid wastes in general can be classified as **rubbish and garbage**

- **Rubbish** refers to the solid waste components that will **not putrefy and that are either combustible** (such as paper, plastic, rubber etc.) or **non-combustible** (such as glass, metal, ceramics, etc.).)
- **Garbage** refers to solid waste that will putrefy (such as fruits, fish, meat, vegetables, etc.).

It is estimated that, on average, 200-600 g of solid waste is generated per person per day in Indian cities. Solid waste includes:

- ✓ Garbage (food waste)
- ✓ Rubbish (paper, plastics, wood, metal, throw-away containers, glass)
- ✓ Demolition products (bricks, masonry, pipes)
- ✓ Sewage treatment residue
- ✓ Dead animals, manure and other discarded material

Environmental Impact of Solid Waste Disposal on Land

- Groundwater contamination by the leachate generated by the waste dump.
- Surface water contamination by the run-off from the waste dump.
- Bad odour, pests, rodents, and wind-blown litter in and around the waste dump.
- generation of inflammable gas (e.g., methane) within the waste dump.
- Bird menace above the waste dump, which affects the flight of aircraft.
- Fires within the waste dump.
- Erosion and stability problems relating to the slopes of the waste dump.
- Epidemics through stray animals.
- Acidity to the surrounding soil and release of greenhouse gas.

WASTE MANAGEMENT

Solid Waste Management (SWM) is the process of collecting and treating solid waste. It also offers solutions for recycling items that are not garbage or trash. As long as people have been living in settlements and residential areas, garbage or solid waste has been an issue.



CONSUMERISM AND THROW-AWAY CULTURE OF SOLID WASTE

- Consumerism of solid waste refers to the direct link between excessive consumer habits and the generation of large amounts of solid waste.
- A "throw-away culture" of solid waste is directly linked to consumerism.
- To buy new products, often designed for single use, results in a significant increase in disposable waste.
- This leads to overflowing landfills and environmental damage.
- Excessive disposal of solid waste led to the **reuse or recycling**.
- People are encouraged to discard items quickly rather than repair or maintain them, contributing to a culture of waste generation

Sources of Solid Waste

1. Municipal Solid Waste (MSW)

Solid waste generated from

- Households (residential areas)
- Commercial sector (hotels and restaurants),
- Institutional buildings (offices and educational institutions) and
- Municipal sector (landscape and tree trimmings, construction and demolition waste and street sweepings)

2. Industrial Waste

- Solid waste generated from industrial activities
- Various industries (such as pulp and paper industry, chemical manufacturing industry etc.) generate a variety of wastes.
- Some are toxic (toxic chemicals) and some are recyclable (paper, plastic and metals).

3. Hospital waste

- Hospital waste is generated during the diagnosis treatment or immunization of human beings or animals
- It includes wastes the form of disposable syringes, needles, bandages, body fluids, contaminated organs of infected persons, etc. which are highly infectious
- Biomedical waste scattered in and around the hospitals invites flies, insects, rodent dogs, etc.
- They play an important role in spreading communicable diseases like plague and rabies

4. Hazardous waste

- Hazardous wastes are wastes that pose a substantial danger either immediately or over a period of time not only to living things (plants, animals and human beings) but also to the environment
- Typical examples are paints, radioactive wastes, pesticides, pathological wastes from hospitals etc.
- They are acutely toxic which can cause cancer and trigger birth defects

5. Agricultural Waste

- Wastes generated from farms, dairies vineyards, fields etc.
- Mainly consist of spoiled food grains, vegetable wastes and animal wastes
- Animal and vegetable wastes can be used to increasing the soil fertility as they contain valuable nutrients

6. Plastic waste

- Plastic has become an intrinsic part of our lives at home, work, and the marketplace.
- This is the reason why we see plastic everywhere in our day-to-day life
- The lifestyle we follow today is "use and throw".
- It is estimated that approximately 56 lakh tonnes of plastic waste is generated in India annually, and most of it ends up on land and in waterways.

7. Construction and Demolition Waste

- Construction and demolition waste is generated whenever any construction/demolition activity takes place, such as building roads, bridges, flyovers, subways, remodelling, etc. It consists mostly of inert and non-biodegradable materials such as concrete, plaster, metal, wood, plastics, etc. A part of this waste comes to the municipal stream.
- Solid waste produced in these places includes steel materials, concrete, wood, plastics, rubber, copper wires, dirt, and glass.

8. Municipal services

- Some of the waste brought about by the municipal services include street cleaning, waste from parks and beaches, waste from wastewater treatment plants, landscaping wastes, and waste from recreational areas including sludge.

9. E- Waste

- E- waste covers any household or business item.
- E- waste contains circuits and electrical components (also known as electronic gadgets).
- Some of them are computers, mobile phones, air conditioners, televisions, etc.
- India ranks among the top 10 countries globally in terms of solid waste generation

CHARACTERISTICS OF MUNICIPAL SOLID WASTE

a) Physical Characteristics

1. Density

The density of solid waste, i.e., its **mass per unit volume (kg/m^3)**. It is a critical factor in the design of an SWM system. Example - design of sanitary landfills, storage, types of collection, and transport vehicles, etc. Usually, it refers to uncompacted waste. Normal compaction equipment can achieve a **reduction in the volume of waste by 75%**. It increases an initial density of 100 kg/m^3 to 400 kg/m^3 . In other words, a waste collection vehicle can haul four times the weight of waste in its compacted state than when it is un-compacted.

2. Moisture Content

Moisture content is defined as the **ratio of the weight of water (wet weight - dry weight) to the total weight of the wet waste**. Loss of moisture occurs when heated to 104°C for 24 h. It is one of the important physical characteristics of solid waste. Moisture increases the weight of solid wastes, and thereby, the **cost of collection and transport**. In addition, the moisture content is a critical determinant in the economic feasibility of waste treatment by incineration. Wastes should be insulated from rainfall or other extraneous water. A typical range of moisture content is 20% to 40%, representing the extremes of wastes in an arid climate and in the wet season of a region of high precipitation.

3. Particle Size

Particle size is an important characteristic of solid waste because of its significance in the design of mechanical separators and shredders. The size and distribution of the components of wastes are also important for the recovery of materials. Waste is separated by mechanical means such as trammel screens and magnetic separators. For example, ferrous items that are of large size may be too heavy to be separated by a magnetic belt so a magnetic separator is needed.

4. Field capacity

Field capacity of MSW is the total amount of **moisture that can be retained in a waste sample** subject to gravitational pull. It is a critical physical characteristic of solid waste because the water in excess of field capacity will form leachate. **Leachate can be a major problem in landfills.** Field capacity varies with the degree of applied pressure and the state of decomposition of the wastes.

5. Permeability

Hydraulic conductivity of compacted waste. It is an important physical property that to a large extent governs the movement of liquids and gases in landfills.

b) Chemical Characteristics

1. Volatile solid content - Volatile matter exhibited an additional loss of weight on ignition at 950°C in a covered crucible.

2. Ash content

- Ash is the amount of material remaining after burning the waste.
- Ash content exhibited a weight of residue after combustion in an open crucible.

3. CHNSO content

- CHNSO content represents Carbon (C), hydrogen (H), nitrogen (N), sulphur (S), and oxygen (O).
- It is critical for optimizing waste-to-energy conversion, waste-to-resource processes, and for meeting environmental regulations.
- C, H, N, S, and O contents were measured through an elemental analyzer (CE Instruments, CHNS-O Model EA-1110, Thermo Electron).

4. Calorific value

Calorific value is the amount of heat generated from the combustion of a unit weight of a substance, expressed as kcal/kg. The energy content of the organic components in MSW can be determined by:

- using a full-scale boiler as a calorimeter,
- using a laboratory bomb calorimeter,
- Calculation, if elemental composition is known.

The calorific value is determined experimentally using a bomb calorimeter in which the heat generated at a constant temperature of 25°C from the combustion of a dry sample is measured.

c) Biological Characteristics

The most important biological characteristic of the organic fraction of solid waste is that it can be converted biologically to gases and relatively inert organic and inorganic solids. The generation of flies is also related to the putrescible nature of the organic materials.

WASTE DISPOSAL METHODS

1. Landfill

A landfill is the site for the disposal of waste materials by burial is called landfill. Modern sanitary landfills are carefully engineered structures designed to separate garbage from nearby water, soil, wildlife, and people. The cavities are dug out of the ground, and then the walls are sealed with layers of clay and coated with plastic to prevent groundwater contamination from wastewater that accumulates. Any waste that is not recycled or reused has to go somewhere and ends up at a landfill site. Landfill sites can be created from a void made by quarrying. It contains both household and commercial rubbish, for example, food, paper, cardboard, or wood. It also contains wastes from construction and commercial businesses, such as soil, concrete, and brick rubble.



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Main components of a landfill are:

1. Bottom liner & side liner
2. Leachate collection system
3. Stormwater drainage
4. Methane collection system
5. Cover/top cap
6. Groundwater monitoring stations

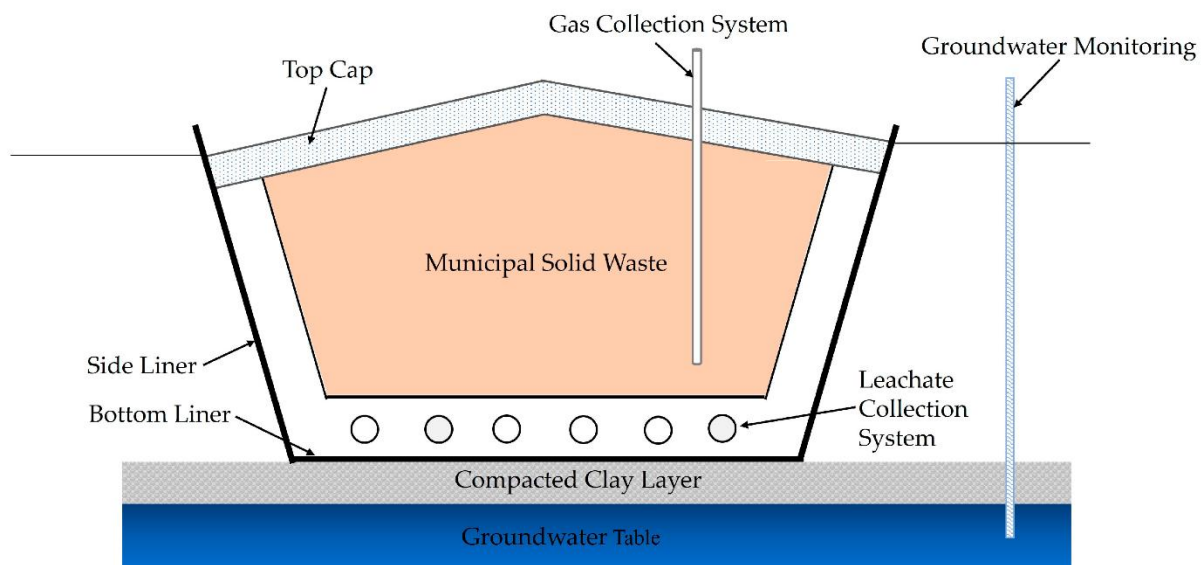
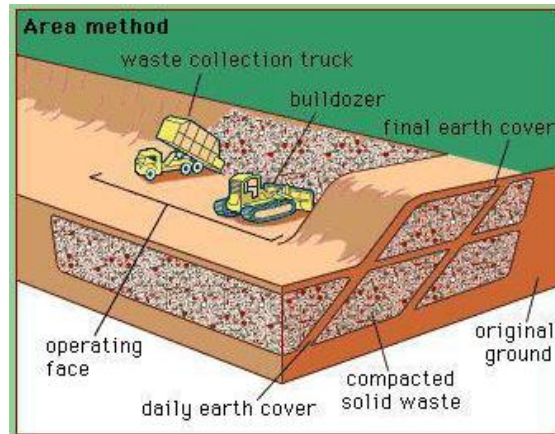


Fig. Cross-section of a landfill

Methods of Landfilling are:

a) Area method

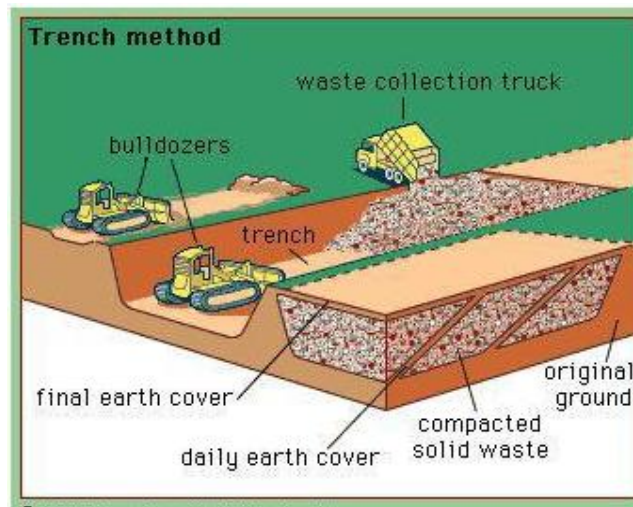
Area landfilling method is also known as **Above Ground Landfill**. Area landfilling is mostly used to dispose urban waste. This method is best suited for flat or gently sloping areas where some land depressions may exist. Area landfilling method involves depositing waste in layers and forming terraces over an identified landfilling area. Area landfilling method generates **excessive leachate**, which is not only difficult to control but will also pollute ground water table. The wastes are spread, compacted and then covered with material which may need to be hauled in from adjacent areas.



b) Trench method

Trench method is also known as **Below Ground Landfill**. This method of landfilling involves excavation of a waste trench. Collected waste is disposed inside excavated trench and is covered-up with a layer of soil. This method is suitable only for small quantity of waste. Trench landfilling method may not be suitable for large quantity of waste because larger trench needs to be excavated with proportionate increase in the cost of disposable.

The trench method is best suited for nearly level land where the water table is not near the surface. Usually, the soil excavated from the trench is used for cover material.



c) Ramp/slope method

Ramp/slope method involves disposal of collected waste along the sides of the existing hill slope. The slope or ramp is sometimes used in combination with the other two methods. The wastes are spread on an existing slope, compacted, and covered. This variation may be suitable for most areas. The cover materials usually come from just ahead of the working face.

d) Valley method

Valley Landfilling Method is also known as Canyon Landfills/depression method. In the valley landfilling method, collected waste is deposited in valleys, low-lying areas, canyons, ravines etc. Used where natural or artificial depressions exist and are used for land filling. Method

depends on the geometry of the site, the hydrology and geology of the site, and access to the site.

e) Cell method

Cell method is a progressive filling and restoration concept that advocates tidy disposal of waste and involves depositing collected waste in a pre-constructed bonded area. Solid waste is spread in thin layers, compacted, and covered daily with soil or alternative materials within a designated, lined area

f) Combination method

More than one of the above listed landfilling methods are employed in combination to dispose of collected waste

Advantages of landfilling

- Simple and economical. No costly plant or equipment is required.
- Separation of different kinds of refuse is not required.
- No residues are evolved and hence no further disposal is required
- Low-lying water-logged areas and odd quarry pits can be easily recovered and put to better use
- The mosquito breeding places are also thus eliminated.

Disadvantages of sanitary landfilling

- Low lying depressions or dumping sites may not always be available
- Evolution of landfill gases
- Highly toxic and polluted leachate generation

2. Incineration

Incineration method is also known as combustion. In this method the waste is burnt at high temperatures to convert into gaseous compounds. Important use of incineration is that it reduces volume of waste by 20-30% of original volume. This process is also known as thermal treatment where solid waste materials are converted by Incinerators into heat, gas, steam and ash.

INCINERATORS

An apparatus for burning waste material, especially industrial waste, at high temperatures until it is reduced to ash. It reduces the volume of the waste by removing combustible matter. Different types of incinerators include fluidized bed incinerator, liquid injection incinerator, multiple hearth incinerator, waste gas flare, direct flame, etc.

Advantages of incineration

- Decreases quantity of waste
- Efficient waste management
- Production of heat and power
- Reduction of pollution
- Incinerators have filters for trapping pollutants
- Saves on transportation of waste
- Provides better control over odour and noise
- Prevents the production of methane gas
- Eliminates harmful germs and chemicals
- Incinerators operate in any weather
- It has a computerized monitoring system
- Occupies a relatively small space
- Uncontaminated groundwater

Disadvantages of incineration

- Expensive
- Pollutes the environment
- Damaging public health
- The possibility of long-term problems
- Ash waste can potentially harm people and the environment

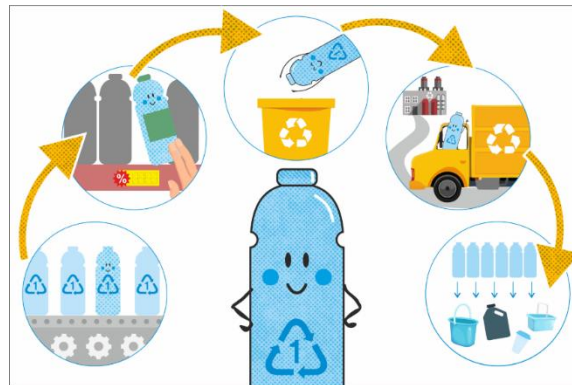
3. Recycling

Recycling is a process of salvaging valuable materials from waste generated so that it can be used again as the same material. Eg: Recycling of food wastes, Recycling of plastics, glass, etc. Recycling can be a primary or a secondary process.

Primary recycling is a process in which the original waste material is made back into the same material. For example, a 'soda can' can be recycled and used as a soda can itself.

In secondary recycling, waste materials are made into different products that may or may not be recyclable. For example, cardboard made from waste newspaper, shredded used tires used in the construction of pavements, and compost from food waste.

For ease of recycling, it is better to segregate the wastes at the source itself. It helps to ensure that waste, which can be processed for recovery of material and energy, does not become co-mixed with undesirable elements.



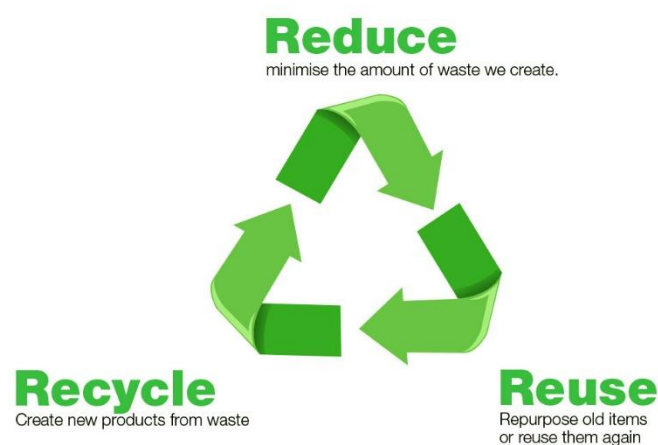
SUSTAINABLE PRACTICES IN WASTE MANAGEMENT

Sustainable waste management system is an efficient way to minimize the hazards or pollution caused by wastes. Sustainable waste management means using material resources efficiently to cut down the number of wastes produced and where waste is generated, dealing with it in a way that actively contributes to the economic, social and environmental goals of sustainable development.

Sustainable waste management practices include:

- Reducing waste
- Reusing materials
- Recycling
- Composting

Reduce, Reuse & Recycle



Reduce

Reduction is the most important strategy of the three R's. It focusses on the source of the waste, or where the waste is originally coming from. Source reduction is carried out when products are designed, manufactured, packaged, and used in a way that limits the amount or toxicity of waste created.

The term reduce means reducing the waste generation at source. Purchase the minimum number of products that is just necessary. This means one has to cut back on unnecessary purchases. This saves not only money and resources but also reduces the waste generated. For example, e-mails minimise paper use. But all this depend upon the attitude and lifestyle of the consumer.

Reuse

The second most important strategy of the three Rs is to Reuse, which is when an item, is cleaned and the materials are used again.

Long lasting goods should be reused over and over with or without minor modification or repair. Bottles, Cans, Cartons etc. can be reused as such. A number of electronic gadgets can be reused again and again after minor repairs. Use shopping bags made of cloth or jute, which can be used over and over again thus increasing its useful lifespan.

Advantages of reuse:

- Reduced disposal, needs and costs.
- Energy and raw material savings.

Recycle

In recycling the generated waste or discarded material that cannot be used again in its original form should be sent back to the industry. There this waste material is broken down so that the industry can use this as a resource for the manufacturing of the product. Thus, recycling occurs in three phases:

- First the waste is sorted and recyclables are collected
- Then the recyclables are used for making the raw materials
- Thereafter these raw materials are used in the production of products

Thus, recycling reduces the amount of waste that is thrown into the community dustbins thereby making the environment cleaner. For example, plastic items are recycled into new plastic products; metal scrap is used to make new metal products. Finally, the waste material that can neither be reused nor recycled must be disposed of in a proper manner with minimum impact to the environment.

Advantages of recycling:

- Saves energy
- Conserves resources

COMPOSTING

Composting is the natural process of 'rotting' or decomposition of organic matter by microorganisms under controlled conditions. It is controlled biological decomposition of organic matter, such as food and yard wastes, into humus. This can be done anaerobically and aerobically. This process takes about 4 to 6 weeks.



► Benefits of composting:

- Provides nutrients to the soil. - Compost improves the quality of soil, and for this reason it is considered as a soil conditioner.
- Increases beneficial soil organisms.
- Protects soil from erosion.
- Assists pollution remediation

Types of composting

1. AEROBIC COMPOSTING

Aerobic composting is the decomposition of organic matter using microorganisms that require oxygen. Oxygen from the air diffuses into the moisture and is taken up by the microbes. As aerobic digestion takes place, the by-products are heat, water, and carbon dioxide (CO₂). The heat produced in aerobic composting is sufficient to kill harmful bacteria and pathogens, as these organisms are not adapted to these environmental conditions. Indore method is an example of aerobic composting. Support the growth of beneficial bacteria species, including psychrophilic, mesophilic, and thermophilic bacteria.

- Psychrophilic – Cold-loving – Temperature <15°C
- Mesophilic – Moderate temperature – 20- 45°C
- Thermophilic bacteria – Heat-loving – Temperature > 45°C



2. ANAEROBIC COMPOSTING

In this decomposition occurs in the absence of oxygen or under the limited supply of oxygen. They produce the intermediate compounds like methane, organic acids, hydrogen sulphide and other substances. These substances have strong odour and some are phytotoxic. It is a low temperature process meaning decomposition takes much longer and doesn't reach sufficient temperatures to safely kill plant pathogens, weed and seeds. The Bangalore method is an example of anaerobic composting.



3. VERMICOMPOSTING

Worm composting is using worms to recycle food scrap and other organic material into a valuable soil amendment called vermicompost. It is a mesophilic process, utilizing microorganisms and earthworms that are active at 10-32°C. The process is faster than composting; because the material passes through the earthworm gut, whereby the resulting earthworm castings ie, rich in nutrients. Earthworms consume various organic wastes and reduce the volume by 40-60%.



TRANSITION TO ZERO WASTE LIFESTYLE

Zero Waste maximizes recycling, minimizes waste, reduces consumption, and ensures that products are made to be reused, repaired or recycled back into nature or the marketplace. Zero waste is aimed at eliminating all discharges to land, water or air that are a threat to planetary, human, animal, or plant health.

Zero waste is a concept for a range of measures aimed at mimicking nature's successful strategies to make our industrial systems work. But moving towards zero waste is a big challenge to humanity. For a successful zero-waste initiative, waste should be considered as a resource.

Benefits of Zero wastes

- Economic benefits: Reduction of waste can, in turn, reduce handling, transport, and treatment costs incurred for solid wastes.
- Faster Progress: A zero-waste strategy improves the production processes and environmental prevention strategies, which can lead to taking larger, more innovative steps.
- Supports sustainability: A zero-waste strategy supports all three of the generally accepted goals of sustainability- economic well-being, environmental protection, and social well-being.
- Improved material flows: A zero-waste strategy would use far fewer new raw materials and send no waste materials to landfills. Any material waste would either return as reusable or recycled materials or would be suitable for use as compost.

CIRCULAR ECONOMY

- An economic model that aims to reduce waste and reuse resources
- In contrast to the 'produce-consume-waste' linear model, a circular economy is based on a restorative or regenerative design of industries, businesses, processes etc.
- This minimizes/eliminates the waste and promotes sustainability throughout the life cycle

- Treating waste as a potential resource and recovering materials from waste streams through recycling might generate revenue and make the value chain self-sustaining

HAZARDOUS AND E-WASTE IDENTIFICATION AND MANAGEMENT

HAZARDOUS WASTE

Hazardous waste includes materials that are toxic, ignitable, corrosive, or reactive, posing significant risks to health and the environment.

1. Listed Waste by US Environmental Protection Agency (US-EPA)

- a) F-List - Non-Specific Source Wastes – Eg. - Spent solvent wastes, Electroplating wastes
- b) K-List - Specific Source Wastes – Eg. - Wood Preservation, Organic Chemical Manufacturing
- c) P-List - Discarded commercial chemical products
- d) U-List - Discarded Commercial Chemical Products

The F-List and K-List pertain to wastes that are generated during manufacturing processes that are carried out at a facility. The P-List and U-List of wastes are usually related to unused, expired, or spilled commercial chemicals.

2. Characterized Waste - Characteristic hazardous wastes are “D” waste

- a) Ignitability (D001)- The ignitability characteristic identifies wastes that can readily catch fire and sustain combustion. Ignitability is not supposed to include all wastes that burn.
- b) Corrosivity (D002) - The corrosivity characteristic identifies wastes that are acidic or alkaline and can readily corrode or dissolve flesh, metal, or other materials.
- c) Reactivity (D003) - The reactivity characteristic identifies wastes that readily explode or undergo violent reactions.
- d) Toxicity (D004) - If the concentration of certain pollutants in the leachate exceeds a standard in the rules, it is a toxic hazardous waste (unless otherwise exempted).

3. Universal Waste

- a) Batteries
- b) Pesticides
- c) Equipment containing mercury
- d) Lamps containing mercury

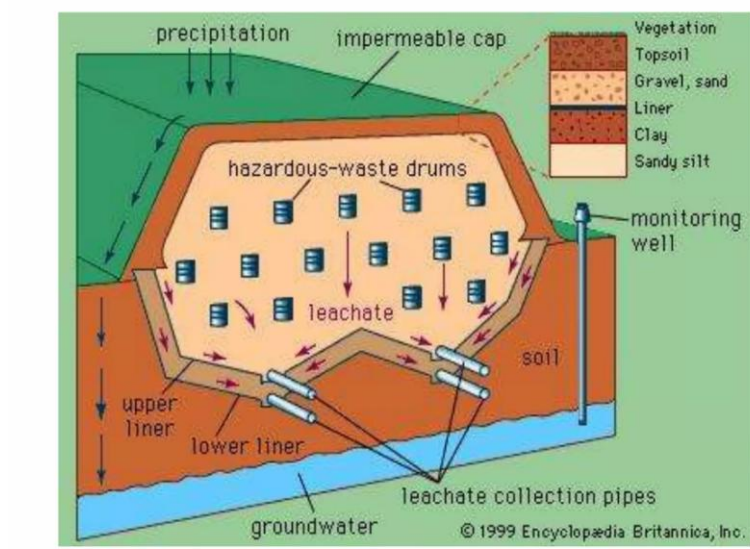
4. Mixed Waste: Radioactive + Hazardous

5. E-Waste: Electrical Waste, Electronic Waste

Hazardous waste management

- Hazardous waste can be treated by chemical, thermal, biological, and physical methods

- Chemical methods include ion exchange, precipitation, oxidation and reduction, and neutralization
- Among thermal methods is high-temperature incineration, which not only can detoxify certain organic wastes but also can destroy them
- Special types of thermal equipment are used for burning waste in either solid, liquid, or sludge form
- Hazardous wastes that are not destroyed by incineration or other chemical processes need to be disposed of properly
- For most such wastes, land disposal is the ultimate destination, although it is not an attractive practice, because of the environmental risk is involved
- A common type of temporary storage impoundment for hazardous liquid waste is an open pit or holding pond, called a lagoon
- New lagoons must be lined with impervious clay soils and flexible membrane liners in order to protect groundwater
- Leachate collection systems must be installed between the liners, and groundwater monitoring wells are required



E-WASTE

- Electronics waste, commonly known as e-scrap or e waste, is the trash we generate from surplus, broken, and obsolete electronic devices
- Electronics contains various toxic and hazardous chemicals and materials that are released into the environment if we do not dispose of them properly
- E-waste or electronics recycling is the process of recovering material from old devices to use in new products

Steps for E-waste management

1. Use 3R technique to minimize the E-waste
2. Identify the e-waste category item
3. Identify the e-waste composition or determine it
4. Identify possible hazardous content in e- waste
5. Identify, whether the e-waste component is hazardous or the entire E-waste item is hazardous

Recycling and waste-to-energy technologies

Main pathways for the conversion of organic waste material to energy

1. **Thermochemical conversion**
2. **Biochemical conversion**
3. **Physicochemical conversion**
4. **Anaerobic digestion**
5. **Biofuel production**

Thermochemical conversion

Thermochemical conversion, characterized by higher temperature and conversion rates, is best suited for lower moisture feedstock. Thermochemical conversion includes three methods incineration, pyrolysis and gasification. The incineration technology is the controlled combustion of waste with the recovery of heat to produce steam which in turn produces power through steam turbines. Pyrolysis and gasification represent refined thermal treatment methods as alternatives to incineration. Pyrolysis is the thermal decomposition of organic material at elevated temperatures in the absence of air. The process, which requires heat, produces a mixture of combustible gases (primarily methane, complex hydrocarbons, hydrogen and carbon monoxide), liquids and solid residues. Thermal decomposition takes place in the presence of a limited amount of oxygen or air. The producer gas which is generated can then be used in either boilers or cleaned up and used in combustion turbine/generators. Gasification is a high-temperature ($>700^{\circ}\text{C}$) thermochemical process that converts carbonaceous materials—such as coal, biomass, or waste—into a clean-burning synthetic gas (syngas).

Biochemical conversion

Biochemical conversion of biomass involves use of bacteria, microorganisms and enzymes to breakdown biomass into gaseous or liquid fuels, such as biogas or bioethanol. Anaerobic digestion (or bio-methanation) and fermentation are the commonly used technologies in Biochemical conversion. Anaerobic digestion is a series of chemical reactions during which organic material is decomposed through the metabolic pathways of naturally occurring microorganisms in an oxygen depleted environment.

Physico-chemical conversion

The physico-chemical technology involves various processes to improve the physical and chemical properties of solid waste. Municipal Solid Waste (MSW) is a poor-quality fuel, and hence pre-processing is necessary to improve its calorific value. The combustible fraction of the waste is converted into high-energy fuel pellets, known as Refuse Derived Fuel (RDF) which may be used in steam generation.

Anaerobic Digestion

Anaerobic Digestion (AD) is the natural biological process that stabilizes organic waste in the absence of air. Anaerobic digestion is a reliable technology for the treatment of wet, organic waste. Organic waste from various sources is biochemically degraded in anaerobic digesters under oxygen-free condition resulting in the production of biogas, which can be used to produce both electricity and heat. Almost any organic material, such as municipal solid waste, animal manure, poultry litter, food wastes, sewage, and industrial wastes, can be used as the feed. Biogas can also be upgraded and used in gas supply networks.

Biofuel production

A variety of fuels can be produced from waste resources including liquid fuels, such as ethanol, methanol, biodiesel and gaseous fuels such as hydrogen and methane. The resource base for biofuel production is composed of a wide variety of forestry and agricultural resources, industrial processing residues, and municipal solid and urban wood residues. Globally, biofuels are most commonly used to power vehicles, heat homes, and for cooking. A schematic diagram explaining the production of Bioethanol.

Ethanol from lignocellulosic biomass is produced mainly via biochemical routes such as pretreatment, enzymatic hydrolysis, and fermentation. Biomass is pretreated to improve the accessibility of enzymes. After pretreatment, biomass undergoes enzymatic hydrolysis for conversion of polysaccharides into monomer sugars, such as glucose and xylose. Subsequently, sugars are fermented to ethanol by the use of different microorganisms.

BIOMEDICAL WASTE

Any waste which is generated during the diagnosis, treatment or immunization of human beings or animals or in research activities pertaining to that or in the production or testing of biological.

The waste generated from medical activities is a severe problem for the human world, even though medical care is vital for our lives and health. Improper management of these wastes has a direct impact on the health of the community. Various communicable diseases can spread through water, sweat, blood, body fluids, and contaminated organs of infected persons. The biomedical waste, scattered in and around the hospitals, invites flies, insects, rodents, cats and dogs that are responsible for the spread of communicable diseases like plague and rabies. The recycling of disposable syringes, needles, IV sets and other articles like glass bottles without proper sterilization is responsible for Hepatitis, HIV, and other viral diseases.

Types of Biomedical Waste

1. Human Anatomical Waste - Human tissues, organs, body parts
2. Animal Waste - Animal tissues, organs, body parts, carcasses, bleeding parts, fluid, blood and experimental animals used in research, waste generated by veterinary hospitals and colleges, discharge from hospitals
3. Microbiology & Biotechnology Waste - Wastes from laboratory cultures, stocks or specimen of live microorganisms, human and animal cell cultures used in research and infectious agents from research and industrial laboratories, wastes from production of biological, toxins and devices used for transfer of cultures
4. Waste Sharps - Needles, syringes, scalpels, blades, glass, etc. that may cause puncture and cuts. This includes both used and unused sharps
5. Discarded Medicine and Cytotoxic drugs - Wastes comprising of outdated, contaminated and discarded medicines
6. Soiled Waste - Items contaminated with body fluids including cotton, dressings, soiled plaster casts, lines, bedding and other materials contaminated with blood
7. Solid Waste (Waste generated from disposable items other than the waste sharps such as tubing, catheters, intravenous sets, etc.
8. Liquid Waste - Waste generated from the laboratory and washing, cleaning, housekeeping and disinfecting activities
9. Incineration Ash - Ash from the incineration of any biomedical waste
10. Chemical Waste - Chemicals used in production of biological, chemicals used in disinfecting, as insecticides, etc.

METHOD OF DISPOSAL OF BIOMEDICAL WASTE/MANAGEMENT OF BIOMEDICAL WASTE

1. Incineration Technology: This is a high-temperature thermal process employing combustion of the waste under controlled condition for converting them into inert material and gases.
2. Autoclaving: An autoclave is a machine that provides a physical method of sterilization by killing bacteria, viruses, and even spores present in the material. In general, an autoclave is run at a temperature of 121° C for at least 30 minutes
3. Microwave irradiation: At a frequency of about 2450 MHz and a wavelength of 12.24 cm, microwaves destroy the majority of microorganisms.
4. Chemical Methods: 1% hypochlorite solution can be used for chemical disinfection.

Colour	Waste	Treatment
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Yellow	Human & Animal anatomical waste /Micro-biology waste and soiled cotton/dressings/linen/beddings etc. Incineration / Deep burial Red Tubing's, Catheters, IV sets.	Autoclaving / Microwaving/Chemical treatment
Red	Tubing's, Catheters, IV sets	Autoclaving / Microwaving/Chemical treatment
Blue/White	Waste sharps - Needles, Syringes, Scalpels, blades etc.	Autoclaving / Microwaving /Chemical treatment & Destruction / Shredding
Black	Discarded medicines/cytotoxic drugs,	Incineration ash, Chemical waste. Disposal in secured landfill



Self-study portions

An overview of:

- CPHEEO guidelines for solid waste management
- Regulations for hazardous waste management in India